

$$\begin{aligned}
C_{\text{dB}\phi_2} \frac{1}{16\pi^2} \rightarrow & \left(\frac{1}{54} \mathbf{g}_1^3 \mathbf{y}_d^{11i2} \text{LF}_{2,1,0} [\mathbf{m}_1, \mathbf{m}_d^{i2}] - \frac{1}{27} \mathbf{g}_1^3 \mathbf{y}_d^{11i2} \text{LF}_{3,1,-1} [\mathbf{m}_1, \mathbf{m}_d^{i2}] + \right. \\
& \frac{1}{54} \mathbf{g}_1^3 \mathbf{y}_d^{11i2} \text{LF}_{4,1,-2} [\mathbf{m}_1, \mathbf{m}_d^{i2}] - \frac{1}{432} \mathbf{g}_1^3 \mathbf{y}_d^{11i2} \text{LF}_{2,1,0} [\mathbf{m}_1, \mathbf{m}_q^{i1}] + \\
& \frac{1}{216} \mathbf{g}_1^3 \mathbf{y}_d^{11i2} \text{LF}_{3,1,-1} [\mathbf{m}_1, \mathbf{m}_q^{i1}] - \frac{1}{432} \mathbf{g}_1^3 \mathbf{y}_d^{11i2} \text{LF}_{4,1,-2} [\mathbf{m}_1, \mathbf{m}_q^{i1}] - \\
& \frac{1}{16} \mathbf{g}_1 \mathbf{g}_2^2 \mathbf{y}_d^{11i2} \text{LF}_{2,1,0} [\mathbf{m}_2, \mathbf{m}_q^{i1}] + \frac{1}{8} \mathbf{g}_1 \mathbf{g}_2^2 \mathbf{y}_d^{11i2} \text{LF}_{3,1,-1} [\mathbf{m}_2, \mathbf{m}_q^{i1}] - \\
& \frac{1}{16} \mathbf{g}_1 \mathbf{g}_2^2 \mathbf{y}_d^{11i2} \text{LF}_{4,1,-2} [\mathbf{m}_2, \mathbf{m}_q^{i1}] + \frac{2}{9} \mathbf{g}_1 \mathbf{g}_3^2 \mathbf{y}_d^{11i2} \text{LF}_{2,1,0} [\mathbf{m}_3, \mathbf{m}_d^{i2}] - \\
& \frac{4}{9} \mathbf{g}_1 \mathbf{g}_3^2 \mathbf{y}_d^{11i2} \text{LF}_{3,1,-1} [\mathbf{m}_3, \mathbf{m}_d^{i2}] + \frac{2}{9} \mathbf{g}_1 \mathbf{g}_3^2 \mathbf{y}_d^{11i2} \text{LF}_{4,1,-2} [\mathbf{m}_3, \mathbf{m}_d^{i2}] - \\
& \frac{1}{9} \mathbf{g}_1 \mathbf{g}_3^2 \mathbf{y}_d^{11i2} \text{LF}_{2,1,0} [\mathbf{m}_3, \mathbf{m}_q^{i1}] + \frac{2}{9} \mathbf{g}_1 \mathbf{g}_3^2 \mathbf{y}_d^{11i2} \text{LF}_{3,1,-1} [\mathbf{m}_3, \mathbf{m}_q^{i1}] - \\
& \frac{1}{9} \mathbf{g}_1 \mathbf{g}_3^2 \mathbf{y}_d^{11i2} \text{LF}_{4,1,-2} [\mathbf{m}_3, \mathbf{m}_q^{i1}] + \frac{1}{12} \mathbf{g}_1 \overline{\mathbf{y}}_d^{\text{pr}} \mathbf{y}_d^{\text{pi}2} \mathbf{y}_d^{i1r} \text{LF}_{2,1,0} [\tilde{\mu}, \mathbf{m}_d^r] - \\
& \frac{1}{24} \mathbf{g}_1 \overline{\mathbf{y}}_d^{\text{pr}} \mathbf{y}_d^{\text{pi}2} \mathbf{y}_d^{i1r} \text{LF}_{3,1,-1} [\tilde{\mu}, \mathbf{m}_d^r] - \frac{1}{24} \mathbf{g}_1 \overline{\mathbf{y}}_d^{\text{pr}} \mathbf{y}_d^{\text{pi}2} \mathbf{y}_d^{i1r} \text{LF}_{4,1,-2} [\tilde{\mu}, \mathbf{m}_d^r] - \\
& \frac{1}{12} \mathbf{g}_1 \overline{\mathbf{y}}_d^{\text{pr}} \mathbf{y}_d^{\text{pi}2} \mathbf{y}_d^{i1r} \text{LF}_{2,1,0} [\tilde{\mu}, \mathbf{m}_q^{\text{p}}] - \frac{1}{12} \mathbf{g}_1 \overline{\mathbf{y}}_d^{\text{pr}} \mathbf{y}_d^{\text{pi}2} \mathbf{y}_d^{i1r} \text{LF}_{3,1,-1} [\tilde{\mu}, \mathbf{m}_q^{\text{p}}] + \\
& \frac{1}{6} \mathbf{g}_1 \overline{\mathbf{y}}_d^{\text{pr}} \mathbf{y}_d^{\text{pi}2} \mathbf{y}_d^{i1r} \text{LF}_{4,1,-2} [\tilde{\mu}, \mathbf{m}_q^{\text{p}}] - \frac{1}{6} \mathbf{g}_1 \mathbf{y}_d^{\text{pi}2} \overline{\mathbf{y}}_u^{\text{pr}} \mathbf{y}_u^{i1r} \text{LF}_{2,1,0} [\tilde{\mu}, \mathbf{m}_u^r] + \\
& \frac{5}{24} \mathbf{g}_1 \mathbf{y}_d^{\text{pi}2} \overline{\mathbf{y}}_u^{\text{pr}} \mathbf{y}_u^{i1r} \text{LF}_{3,1,-1} [\tilde{\mu}, \mathbf{m}_u^r] - \frac{1}{24} \mathbf{g}_1 \mathbf{y}_d^{\text{pi}2} \overline{\mathbf{y}}_u^{\text{pr}} \mathbf{y}_u^{i1r} \text{LF}_{4,1,-2} [\tilde{\mu}, \mathbf{m}_u^r] + \\
& \frac{1}{216} \mathbf{m}_1 \mathbf{g}_1^3 \mathbf{a}_d^{11i2} \text{LF}_{2,1,1,0} [\mathbf{m}_1, \mathbf{m}_d^{i2}, \mathbf{m}_q^{i1}] + \frac{1}{144} \mathbf{m}_1 \mathbf{g}_1^3 \mathbf{a}_d^{11i2} \text{LF}_{2,2,1,-1} [\mathbf{m}_1, \mathbf{m}_d^{i2}, \mathbf{m}_q^{i1}] - \\
& \frac{1}{216} \mathbf{m}_1 \mathbf{g}_1^3 \mathbf{a}_d^{11i2} \text{LF}_{3,1,1,-1} [\mathbf{m}_1, \mathbf{m}_d^{i2}, \mathbf{m}_q^{i1}] + \frac{1}{18} \mathbf{g}_1^3 \mathbf{y}_d^{11i2} \text{LF}_{1,1,1,0} [\mathbf{m}_1, \mathbf{m}_d^{i2}, \tilde{\mu}] - \\
& \frac{1}{9} \mathbf{g}_1^3 \mathbf{y}_d^{11i2} \text{LF}_{2,1,1,-1} [\mathbf{m}_1, \mathbf{m}_d^{i2}, \tilde{\mu}] + \frac{1}{18} \mathbf{g}_1^3 \mathbf{y}_d^{11i2} \text{LF}_{3,1,1,-2} [\mathbf{m}_1, \mathbf{m}_d^{i2}, \tilde{\mu}] - \\
& \frac{1}{144} \mathbf{m}_1 \mathbf{g}_1^3 \mathbf{a}_d^{11i2} \text{LF}_{2,2,1,-1} [\mathbf{m}_1, \mathbf{m}_q^{i1}, \mathbf{m}_d^{i2}] - \frac{1}{72} \mathbf{g}_1^3 \mathbf{y}_d^{11i2} \text{LF}_{1,1,1,0} [\mathbf{m}_1, \mathbf{m}_q^{i1}, \tilde{\mu}] + \\
& \frac{1}{36} \mathbf{g}_1^3 \mathbf{y}_d^{11i2} \text{LF}_{2,1,1,-1} [\mathbf{m}_1, \mathbf{m}_q^{i1}, \tilde{\mu}] - \frac{1}{72} \mathbf{g}_1^3 \mathbf{y}_d^{11i2} \text{LF}_{3,1,1,-2} [\mathbf{m}_1, \mathbf{m}_q^{i1}, \tilde{\mu}] + \\
& \frac{1}{72} \mathbf{g}_1^3 \mathbf{y}_d^{11i2} \text{LF}_{2,2,1,-2} [\mathbf{m}_1, \tilde{\mu}, \mathbf{m}_d^{i2}] + \frac{1}{144} \mathbf{g}_1^3 \mathbf{y}_d^{11i2} \text{LF}_{2,2,1,-2} [\mathbf{m}_1, \tilde{\mu}, \mathbf{m}_q^{i1}] - \\
& \frac{1}{8} \mathbf{g}_1 \mathbf{g}_2^2 \mathbf{y}_d^{11i2} \text{LF}_{1,1,1,0} [\mathbf{m}_2, \mathbf{m}_q^{i1}, \tilde{\mu}] + \frac{1}{4} \mathbf{g}_1 \mathbf{g}_2^2 \mathbf{y}_d^{11i2} \text{LF}_{2,1,1,-1} [\mathbf{m}_2, \mathbf{m}_q^{i1}, \tilde{\mu}] - \\
& \frac{1}{8} \mathbf{g}_1 \mathbf{g}_2^2 \mathbf{y}_d^{11i2} \text{LF}_{3,1,1,-2} [\mathbf{m}_2, \mathbf{m}_q^{i1}, \tilde{\mu}] + \frac{1}{16} \mathbf{g}_1 \mathbf{g}_2^2 \mathbf{y}_d^{11i2} \text{LF}_{2,2,1,-2} [\mathbf{m}_2, \tilde{\mu}, \mathbf{m}_q^{i1}] - \\
& \frac{1}{9} \mathbf{g}_1 \mathbf{m}_3 \mathbf{g}_3^2 \mathbf{a}_d^{11i2} \text{LF}_{2,1,1,0} [\mathbf{m}_3, \mathbf{m}_d^{i2}, \mathbf{m}_q^{i1}] - \frac{1}{6} \mathbf{g}_1 \mathbf{m}_3 \mathbf{g}_3^2 \mathbf{a}_d^{11i2} \text{LF}_{2,2,1,-1} [\mathbf{m}_3, \mathbf{m}_d^{i2}, \mathbf{m}_q^{i1}] + \\
& \frac{1}{9} \mathbf{g}_1 \mathbf{m}_3 \mathbf{g}_3^2 \mathbf{a}_d^{11i2} \text{LF}_{3,1,1,-1} [\mathbf{m}_3, \mathbf{m}_d^{i2}, \mathbf{m}_q^{i1}] + \frac{1}{6} \mathbf{g}_1 \mathbf{m}_3 \mathbf{g}_3^2 \mathbf{a}_d^{11i2} \text{LF}_{2,2,1,-1} [\mathbf{m}_3, \mathbf{m}_q^{i1}, \mathbf{m}_d^{i2}] - \\
& \frac{1}{16} \mathbf{g}_1 \tilde{\mu}^2 \mathbf{y}_d^{\text{pi}2} \overline{\mathbf{y}}_u^{\text{pr}} \mathbf{y}_u^{i1r} \text{LF}_{2,2,1,-1} [\mathbf{m}_q^{\text{p}}, \tilde{\mu}, \mathbf{m}_u^r] + \\
& \frac{1}{16} \mathbf{g}_1 \tilde{\mu}^2 \mathbf{y}_d^{\text{pi}2} \overline{\mathbf{y}}_u^{\text{pr}} \mathbf{y}_u^{i1r} \text{LF}_{2,2,1,-1} [\mathbf{m}_u^r, \tilde{\mu}, \mathbf{m}_q^{\text{p}}] - \frac{1}{36} \mathbf{g}_1^3 \mathbf{y}_d^{11i2} \text{LF}_{2,1,1,-1} [\tilde{\mu}, \mathbf{m}_1, \mathbf{m}_d^{i2}] - \\
& \frac{1}{36} \mathbf{g}_1^3 \mathbf{y}_d^{11i2} \text{LF}_{3,1,1,-2} [\tilde{\mu}, \mathbf{m}_1, \mathbf{m}_d^{i2}] - \frac{1}{72} \mathbf{g}_1^3 \mathbf{y}_d^{11i2} \text{LF}_{2,1,1,-1} [\tilde{\mu}, \mathbf{m}_1, \mathbf{m}_q^{i1}] + \\
& \frac{1}{36} \mathbf{g}_1^3 \mathbf{y}_d^{11i2} \text{LF}_{3,1,1,-2} [\tilde{\mu}, \mathbf{m}_1, \mathbf{m}_q^{i1}] - \frac{1}{8} \mathbf{g}_1 \mathbf{g}_2^2 \mathbf{y}_d^{11i2} \text{LF}_{2,1,1,-1} [\tilde{\mu}, \mathbf{m}_2, \mathbf{m}_q^{i1}] + \\
& \frac{1}{4} \mathbf{g}_1 \mathbf{g}_2^2 \mathbf{y}_d^{11i2} \text{LF}_{3,1,1,-2} [\tilde{\mu}, \mathbf{m}_2, \mathbf{m}_q^{i1}] + \frac{5}{24} \mathbf{g}_1 \tilde{\mu}^2 \mathbf{y}_d^{\text{pi}2} \overline{\mathbf{y}}_u^{\text{pr}} \mathbf{y}_u^{i1r} \text{LF}_{2,1,1,0} [\tilde{\mu}, \mathbf{m}_q^{\text{p}}, \mathbf{m}_u^r] + \\
& \left. \frac{1}{24} \mathbf{g}_1 \tilde{\mu}^2 \mathbf{y}_d^{\text{pi}2} \overline{\mathbf{y}}_u^{\text{pr}} \mathbf{y}_u^{i1r} \text{LF}_{3,1,1,-1} [\tilde{\mu}, \mathbf{m}_q^{\text{p}}, \mathbf{m}_u^r] \right)
\end{aligned}$$